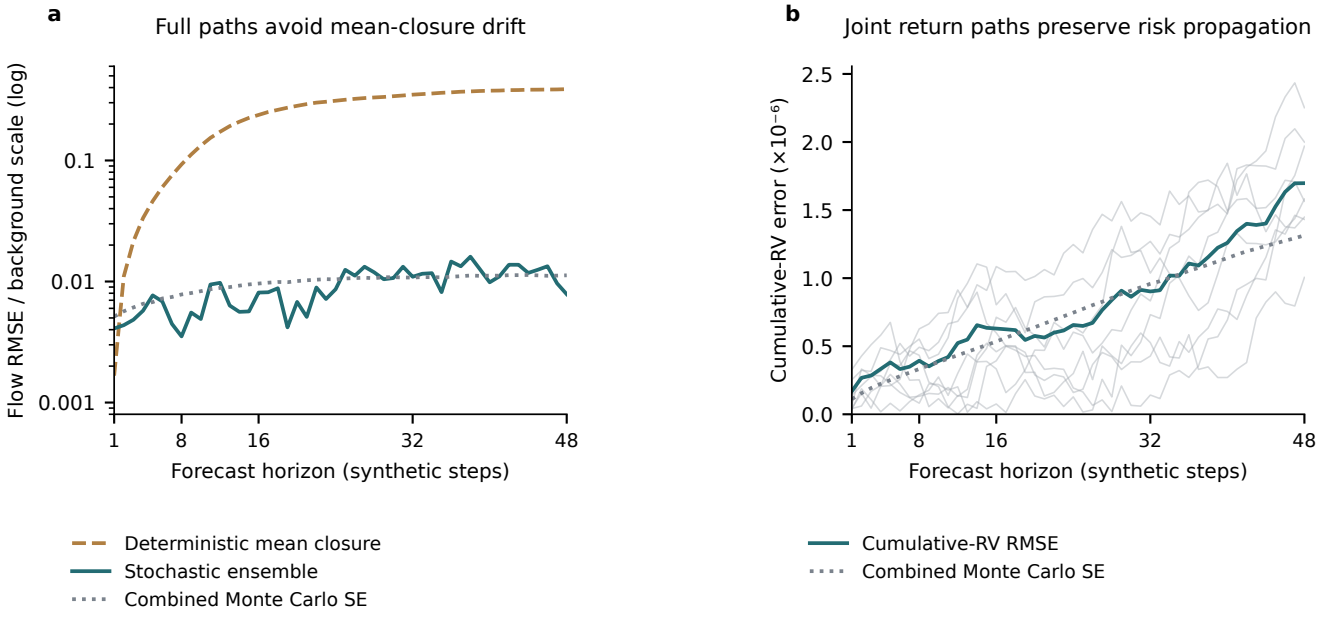
a
Full paths avoid mean-closure drift
Flow RMSE / background scale (log)
0.1
0.01
0.001
1
8
16
32
48
Forecast horizon (synthetic steps)
Deterministic mean closure
Stochastic ensemble
Combined Monte Carlo SE
b
Joint return paths preserve risk propagation
Cumulative-RV error (×10⁻⁶)
2.5
2.0
1.5
1.0
0.5
0.0
1
8
16
32
48
Forecast horizon (synthetic steps)
Cumulative-RV RMSE
Combined Monte Carlo SE